Passive Precision Time over BLE
V14 Service and Profile Release Candidate

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Abstract

Passive Precision Time over BLE (PToB) transfers shared time to heterogeneous Bluetooth Low Energy receivers. Periodic advertisements identify physical transmissions; later corrections let receivers discipline local clocks without a transaction for each timestamp. This release candidate pins the implemented BLE wire profile: G7V4 grandmaster control, GR3 root and RC7/profile 5 relay timing, JOIN5 and revision 8 relay admission, and coarse civil-time metadata. It defines when a source may serve, when a HOT successor may take over, and when a receiver must withdraw time. Byte compatibility, physical failover, and calibrated output accuracy are separate review gates.

1 Purpose, scope and roles

The service lets an endpoint identify a time source, maintain a local clock between observations, report provenance and withdraw unavailable time. Receivers may range from coarse callback consumers to hardware timestamping instruments. Packet resolution alone makes no accuracy promise.

MUST, MUST NOT and SHOULD express candidate requirements. The reference is the T096 0.15.326 source snapshot identified in the edition manifest. Its GM profile is `GM7v4+S2+C1+TZ1`: G7V4 wire, ordinary broadcast following, full mesh birth and authenticated time-zone metadata. G1 names the timing engine; GR3 names the selected root wire. V14 identifies the service edition, independent of wire revision and firmware version [1, 4, 3].

Role	Function and authority
ACTIVE GM	Independently qualified source currently originating mesh time.
HOT GM	Designated successor with qualified own source and a usable same-scope shadow clock; conditionally prepared for takeover.
COLD / following GM	Receives mesh time and maintains its own source; has no fast-takeover entitlement.
Relay clock (RC)	Follows one authenticated parent and regenerates downstream timing; receives no GM election or HOT authority.
Client / leaf	Consumes an admitted source. Capture capability determines its timing tier; a parent constraint remains binding.

A device's role, functional clock availability and precision quality MUST remain separately observable. Source transitions fence authority, captures and maps before replacement service. This BLE candidate leaves LoRa/RC8, connected RC9 bridging, and USB or wired time delivery to separate contracts.

The time scale is a POSIX UTC estimate with an authenticated source-coordinate mapping. Leap/TAI handling and wired PTP/gPTP translation need separate scale, delay and uncertainty contracts. Civil time never replaces UTC event identity.

2 GM7 election, following and failover

GM control uses authenticated extended advertisements. ANNOUNCE (ANN) names the ACTIVE source and its nominee, if any. A following GM listens to the selected source's GR3 train while its own GNSS/PPS continues. It reports independent eligibility and actual shadow preparation in STATUS; a qualified nominee echoes HOT in ordinary STATUS. GM following needs no JOIN, GATT, nonce or ANN-based receive lease. An ACTIVE GM still uses JOIN for ordinary clients [1].

Eligibility precedes ranking. The reference source requires 60 seconds of continuous independent GNSS/PPS qualification. Ordinary initial election then requires 30 seconds of quiet observation plus deterministic backoff up to 500 ms: a healthy cold start normally takes about 90 seconds plus backoff. A GM lacking its own qualified source may follow time but MUST NOT originate it. Rank eligible candidates lexicographically by priority1, clock class, accuracy, variance, priority2 and stable identity, with boot identity resolving incarnation ties. Preference alone is never HOT confirmation.

HOT readiness requires a current same-scope delivered shadow map and fit, live correction/source evidence, qualified own GNSS/PPS and a physical phase enclosure within the 2 ms scheduling policy. Strict precision LOCK is a separate assessment. Nomination, diagnostics and displayed RMS cannot substitute for those readiness predicates.

Transition	Required behavior
Selected GM silence	Only the designated, still-qualified HOT uses the fast path: trigger at 2.5 s after last advancing selected-GM communication; target first completed G1 within 3 s of that endpoint. Actual capture, drain, source and reservation gates still apply.
Explicit YIELD	Stop-before-start at the immutable boundary, 750 ms after the original source sample; translate it through the retained physical map. Three copies 250 ms apart cannot move it.
WITHDRAW / source fault	Stop positive output immediately; attempt three immutable withdrawal copies within their original 1 s negative lifetime. A successor still needs its own authority and source readiness.
No usable HOT	An independently qualified COLD GM uses ordinary 30 s quiet election plus backoff. Held remote time cannot create source eligibility.
Fresh selected contact	Advancing ANN, authenticated self-ACTIVE ROLE or direct timing header inhibits silence-only takeover. Duplicate/reordered traffic does not renew liveness.

GM following continues across control loss. Actual matched corrections, fit freshness and original source expiry bound delivered time. The receiver converts the correction's remaining grant to its captured RX coordinate, rounding down. An advancing selected direct header can establish liveness before a match is usable; it cannot train the clock. A remembered listening identity grants no map. The 3 s objective requires physical qualification.

Mesh lineage and convergence

At first valid serving reservation, sample full 64-bit functional UTC microseconds as mesh birth and record the founder. Commit ACTIVE only after successful bounded ANN transmission; track actual RF completion separately. A possibly exposed claim retains its lineage even if completion fails. Successors inherit birth/founder and advance the leader term; boot, source epoch, capture generation and train remain separate scopes [2].

When ACTIVE meshes meet, earlier birth wins; exact-birth ties use founder, then term/identity/boot ordering. The newer mesh stops positive output before following. The older ACTIVE keeps its service and nominee. Disconnected partitions can serve separate independently eligible sources; convergence does not promise consensus across absent links. Birth is a functional ordering value, not proof of calibrated UTC.

3 Periodic timing and relay regeneration

Timing uses LE periodic advertising, with complete application records at a nominal 62.5 ms interval (16 Hz). Acquisition uses extended-advertising discovery; synchronization is distinct from connecting. This uses periodic advertising without responses [9]. An application record may span controller reports or link-layer fragments, so adapters MUST deliver bounded whole records and preserve their event identity.

An original publication identifies a physical transmission without predicting its own future TX timestamp. A later record carries zero, one or two corrections naming earlier publications, periodic events and physical data channels (0–36). The receiver matches a correction to the saved physical RX capture of that exact original. Callback arrival, follow-up RX and worker processing time are different events and MUST NOT replace that capture.

The association scope includes authenticated group, root identity/boot/source epoch, immediate sender identity/boot, train incarnation, capture epoch, publication, periodic event, channel and adapter/PHY context. Missing originals, ambiguous copies, replay, nonpreceding corrections and retired scope are refused. Counter wrap is valid only within its bounded live scope; overflow cannot alias an earlier event.

Let $M_g(x)$ map local counter x to source time in immutable generation g. A completed pair supplies the anchor

$$M_g(x_{\mathrm{RX}}) = T_{\mathrm{TX}} + d,$$

where d is the declared path/reference treatment with its uncertainty. A new map becomes usable only after correction processing completes. Earlier reads or witness events retain the map then available; later evidence cannot improve past output.

Boundary clocks and paths

An RC disciplines its own counter and originates a downstream train. Each correction uses the map effective at its own TX event and preserves that generation. It MUST NOT present the parent's physical event as its own. Root identity, boot, source epoch, original issuance/expiry, UTC mapping and uncertainty remain provenance; sender boot, train and capture context describe the new transmitter.

RC7 authenticates a dynamic bounded path of root and at most two relay transmitters, including identities, boots, trains, capture epochs and slots. It rejects loops, incompatible scope and invalid parent/depth relationships. A leaf does not acquire relay or GM authority through admission. Root position zero and nominal RC positions at 15.625, 31.250 and 46.875 ms are scheduling policy, not PAwR response slots or proof of actual controller placement. Missed preparation advances by whole frames. Port-specific TX estimates require an explicit timing tier and uncertainty [3].

Validity and continuity

Functional availability, local discipline age, inherited age, independent source permission, admission/demand and precision quality are separate. T096 starts same-parent renewal at 240 s for direct and relay bindings, with at most a 360 s lease. The root source envelope is at most 1.65 s. Each consumer also applies its actual correction, fitter and profile expiry limits.

Holdover needs a declared model, elapsed age and supported error budget. The T096 RC's local BLE HOLD estimate has unknown bounds and no relay, USB observation, or source authority [7]. A receiver may keep acquiring the same authenticated direct train during a permission gap while withdrawing timing output; a context change requires fresh admission [8]. Reboot, source/counter epoch change, revocation or arithmetic fault fences affected output. Within a continuous valid scope, delivered time is monotonic; reacquisition discontinuities are explicit.

4 BLE wire profile and authentication

Lengths below include the four-byte manufacturer-specific AD framing where present. A framed record starts with length (record bytes minus one), type `ff`, and development company value `ff ff`. JOIN has no AD prefix. Relay control uses an AD-shaped record over the connected exchange. Numeric offsets are zero-based; half-open slices are written $[a : b)$ [3, 4].

Record	Selected wire / size	BLE bearer and discriminator
Root timing	GR3 v3; 190 bytes	Periodic AD: `GR`, revision 3 at bytes 4–6.
Relay timing	RC7 v7/profile 5; 190 bytes	Periodic AD: `RC`, revision 7 at 4–6; profile at 172.
GM control	G7V4 rev. 4; 182 body / 186 AD	Extended AD: ANN, STATUS, WITHDRAW; YIELD is an ANN flag.
GM observations	G7V4 rev. 4; ROLE 168 body / 172 AD, GNSS 156 body / 160 AD	Extended AD: descriptive ROLE and GNSS; self-ACTIVE ROLE can select a listening identity.
Direct admission	JOIN5; 168 bytes	Connected GATT record; decimal 14 at byte 0, revision 5 at 2.
Relay control	RQ rev. 8/profile 5; 216 bytes	Connected GATT request/response; `RQ`, revision at 6, profile at 8.
Relay capability	RQ rev. 8/profile 5; 148 bytes	Extended AD; same header, kind 1 at byte 7.

All tags use the first 16 bytes of HMAC-SHA256 with a 32-byte group PSK, over an eight-byte ASCII domain (no terminator), then context, then the authenticated input. Context is byte `01`, little-endian u64 group ID and little-endian u32 key epoch. Precision and JOIN fields are little-endian; GM integers are big-endian except TZ. Signed integers use two's complement.

Authentication	Domain	Input and tag position
GR3 origin / packet	`PTOBG3O3` / `PTOBG3R3`	AD $[14 : 74)$, tag at 74 / AD $[0 : 174)$, tag at 174.
RC7 origin / path	`RC7ORIGN` / `RC7PATHS`	AD $[12 : 72)$, tag at 72 / canonical admitted path, yielding a 16-byte path MAC.
RC7 packet	`RC7PACKT`	Path MAC followed by AD $[0 : 174)$; tag at 174.
GM / JOIN5	`PTOBTZ04` / `PTOBTZJ5`	GM body before its tag / record $[0 : 152)$, tag at 152.
RQ control / cap.	`PTOBTZR8` / `PTOBTZC8`	Record $[0 : 200)$, tag at 200 / AD $[0 : 132)$, tag at 132.

GR3 correction slots start at 136 and 155, each 19 bytes: u32 publication, u16 event, u8 channel, u64 TX in 16 MHz ticks and u32 generation. RC7 slots start at 118 and 145, each 27 bytes: u32 publication, u16 event, u8 channel, u32 generation, u32 cumulative age in microseconds, u64 TX and u32 bound in nanoseconds. Correction count is at byte 7 in both; unused slots are zero. Valid precision bytes do not admit a source using an incompatible coarse revision.

The RC7 path MAC input is 89 bytes: u32 source epoch, u8 depth, and three 28-byte nodes. Each node is u64 ID, u64 boot, u32 train, u32 capture epoch, u8 slot, u8 role and two zero bytes. Unused nodes are all zero. The admitted path is required to verify the timing packet.

5 G7V4 grandmaster record

Every G7V4 body begins with magic at 0–3, revision 4 at 4, kind at 5, exact big-endian u16 body length at 6–7, flags at 8–9, zero at 10–11, u64 group ID at 12–19 and u32 key epoch at 20–23. Body offsets below exclude the AD prefix; add four for an AD offset. The entire pre-tag body, including its header, is authenticated [3, 4].

Kind	Body / tag start	Effect
1 ANN	182 bytes / 166	ACTIVE lineage and nominee; may carry YIELD.
2 STATUS	182 bytes / 166	Eligibility and preparation; nominated member can confirm HOT.
3 WITHDRAW	182 bytes / 166	Terminal notice for that serving claim. No positive source grant.
6 ROLE	168 bytes / 152	Reports role and blockers. Self-ACTIVE can select identity and lineage.
7 GNSS	156 bytes / 140	Reports local source quality and known/unknown state.

ANN, STATUS and WITHDRAW body

Body bytes	Fields in byte order
24–55	Four u64: sender ID, sender boot, sequence, sent UTC in microseconds.
56–75	Five u32: observation maximum age and source-valid interval (microseconds), source, capture and train generations.
76–81	Four u8: priority1, clock class, accuracy, priority2; u16 variance.
82–121	Five u64: mesh birth UTC in microseconds, founder, term, ACTIVE ID, ACTIVE boot.
122–145	Three u64: HOT nominee ID, nominee boot, YIELD UTC in microseconds.
146–161	u32 shadow generation, u64 sender monotonic microseconds, u32 timing epoch.
162–181	Four TZ bytes (little-endian zone and offset), then 16-byte tag.

Flag masks are 1 ELIGIBLE, 2 SHADOW, 4 HOT, 8 YIELD and 16 CAPABLE. ANN and WITHDRAW must name the sending ACTIVE incarnation and carry full lineage. STATUS is the only kind that can set CAPABLE or echo HOT; a HOT echo identifies the sender's own boot and requires eligibility and shadow readiness. YIELD is valid only on ANN, with a named nominee and a stop boundary exactly 750 ms after the original UTC sample. WITHDRAW has zero flags. Maximum observation ages are 2.75 s for ANN, 4.5 s for STATUS and 1 s for WITHDRAW. Only ANN carries a positive source-valid field (at most 1.5 s); it does not grant GM receive permission.

ROLE and GNSS body

Both kinds have zero header flags and no TZ field. Bytes 24–95 contain nine u64 values: sender, boot, sequence, sample monotonic microseconds, birth UTC microseconds, founder, term, ACTIVE ID and boot. Timing epoch and lifetime in microseconds are u32 at 96 and 100. ROLE then carries nominee ID/boot at 104–119, generations at 120–135, role/known/blockers at 136–139 and three sampled durations at 140–151. GNSS carries generations at 104–115, known/value masks at 116–119 and quality, ages and UTC error at 120–139. Observation limits are 4.5 s and 30 s, respectively; neither kind grants a timing map or HOT authority.

6 Admission, receiver behavior and trust

Receivers MUST discover local meshes, keep trusted credentials separate from runtime mesh selection and authenticate admission afresh after context changes. A valid shared-key tag authenticates group membership and integrity. Any key holder can originate statements, so the tag provides no individual attestation or keyless verification. Payloads are readable on air.

Ordinary clients use direct JOIN or relay-local admission; GMs follow broadcast-selected timing. The T096 GATT transport writes complete records without response and requires an ATT MTU of at least 226 for group control. JOIN binds identities, boots, fresh nonces and forward sequences. The exchange state expires after 5 s; source-bearing extraction separately requires RTT below 250 ms. Coarse midpoint uncertainty includes half the RTT, the independent source bound and a drift/quantization margin. Remaining permission subtracts the full RTT plus those guards. The replay anchor MUST NOT train the precision estimator. Relay admission binds the path, parent boot, challenge, nonce, publication floor and bounded source context [3].

Demand and output availability are independent: an authenticated request can start acquisition while timing is unavailable. A not-ready response grants no timing. Renewing same-scope context preserves still-valid captures/maps when resources allow but MUST NOT reset source age or resurrect expired output. Ordinary source advertisements and current usable source evidence, not a long demand lease, determine whether an ACTIVE GM can admit clients.

Receive procedure

1. Reassemble a bounded whole record with transport and capture identity. Check exact length, discriminator, version/profile, kind and canonical reserved fields.

2. Verify group/key context and all applicable packet, origin and path tags. Bind the record to the selected sender, boot and admitted source/path scope before committing state.

3. Apply replay, sequence/publication floors, terminal/term fences and the original receipt/source deadlines. Reject foreign, duplicate, conflicting, expired or incompatible scope without renewing state.

4. Save a valid original's physical RX capture. Accept a later correction only for an unambiguous preceding original in that scope; check timestamp conversion and arithmetic bounds.

5. Publish a new immutable map only after processing completes. Report functional availability, source provenance, cumulative age and known or unknown uncertainty separately. Preserve refusals and unavailable opportunities.

For ordinary GM7 broadcasts, original receipt time bounds local observation lifetime; polling, duplicate copies and delayed processing cannot extend it. Authentication and retained floors cannot prove the age of a previously unseen recording or advancing recorded stream. This first-contact replay limitation is explicit, including after eviction from bounded peer memory. Durable replay across arbitrary state loss, key rotation/revocation, compromise recovery, asymmetric public verification and delay-attack resistance need production security-profile work.

Capture callbacks use bounded work and storage. Report queue overflow without fabricating captures. Serving transitions require valid controller reservations, actual drain and guards against duplicate output. Ending diagnostic collection MUST NOT stop healthy service. Explicit stop, fault containment and update drain remain bounded.

7 Time zones and daylight-saving support

The serving GM publishes one atomic pair in authenticated coarse metadata. TZ starts at GM body byte 162 (AD byte 166), JOIN byte 148, relay control byte 196 and relay capability byte 128. These are metadata offsets; the tag follows the four-byte pair.

Bytes	Field	Meaning
0–1	`tz_id`, u16 LE	Stable append-only PToB identifier mapped to an IANA zone name.
2–3	`tz_offset_min`, i16 LE	Current effective whole minutes east of UTC; valid range -840 to +840.

ID 0 is unknown, ID 1 is UTC (offset zero), and ID 2 is America/Chicago. The canonical unknown pair is `00 00 00 80`, representing (0, -32768). Chicago CDT is `02 00 d4 fe`, or (2, -300); CST is (2, -360). IDs identify regions, not seasonal abbreviations, and are PToB assignments rather than IANA numeric identifiers [4, 5].

Receivers format local civil time as UTC plus offset×60 seconds, with checked signed arithmetic. They use the transmitted effective offset and need no zone database or DST calculation. An unfamiliar nonzero ID with a valid offset remains usable and forwards unchanged. Invalid metadata normalizes to unknown without invalidating otherwise valid authenticated UTC; malformed framing or authentication failure rejects the packet. Unknown/expired metadata falls back to UTC.

Selection, ordering and propagation

The GM may use automatic location-based selection, a manually selected zone/calendar, or an explicit fixed offset. The reference automatic selector uses its own fresh GNSS fixes, conservative region coverage and a finite calendar validity window. Ambiguous/stale location permits only the declared bounded hold; an expired calendar produces unknown immediately. Standby GMs may prepare their own zone; on takeover the new ACTIVE publishes its own selected pair, including unknown if unavailable. Geographic coverage and calendar support are implementation capabilities, not universal protocol guarantees [6].

An RC learns the pair only from its selected authenticated parent and forwards it unchanged in coarse capability/admission records. Direct updates require ordered selected-GM ANN with current root/boot/epoch scope and replay checks. Initial JOIN metadata cannot overwrite a later accepted ANN. Relay changes use the existing monotonic admission challenge: a higher token can change the pair; an equal token renews only an identical pair. Older or equal-token conflicting updates are refused. No TZ counter or transition schedule is added.

TZ has a local presence lease of three nominal parent advertising periods: 750 ms for a 250 ms GM cadence, 300 ms for a 100 ms RC cadence. Matching authenticated parent advertisements may renew that lease independently of precision corrections, lock and map expiry. Each hop renews its own presence lease. Polling cannot renew it. After expiry the live value is unknown; the last accepted pair may support matching-presence renewal. An explicit unknown update clears that pair, and parent/path replacement fences the old context.

Receivers must continue receiving coarse updates after timing synchronization. Precision presence can renew a cached pair but carries no changed pair. A DST transition changes the effective offset on fresh coarse reception; simultaneous scheduled display changes are not promised. Repeated local hours retain distinct UTC event identities.

TZ MUST NOT alter precision G1/GR3/RC7 payloads, UTC mappings, clock discipline, PPS, source/mesh epochs, replay comparisons, USB UTC or chrony inputs. The precision field `utc_offset_ns` maps source coordinates to UTC and MUST NOT be repurposed for civil-time offset. TZ grants no timing permission. Other bearers require their own explicit metadata contract.

8 Conformance targets and proposed SIG work

Conformance is layered: byte interoperability, state/authority correctness, radio-event association, functional availability and independently measured precision require separate results. No codec, model, native build, fitted residual or successful radio request establishes all of them.

Area	Required verification
Wire and bearers	Exact versions/lengths/tags, canonical padding, complete reassembly, malformed/refused vectors, transport/discovery binding and cross-vendor decoding.
GM7 authority	Independent qualification; initial election; actual HOT confirmation; source/control loss; immutable YIELD/WITHDRAW; physical stop-before-start; conditional 3 s first-G1 target; no-HOT fallback; partition merge.
Timing and relays	Unique TX/RX originals and corrections; capture/scope transitions, replay and wrap; causal map activation; original expiry; one- and two-relay paths; actual controller placement and loss recovery.
TZ and DST	Signed/fractional-hour offsets, unknown/future IDs, ordered direct/relay updates, withdrawal and exact lease expiry, parent/GM replacement, repeated/skipped local hours and unchanged UTC/precision bytes.
Precision and resources	Named output/reference points, calibration and uncertainty, bias and total RMS, tails, unavailable/missing counts, drift/holdover, coexistence, energy and sustained operation.
Security and lifecycle	Current admission, key/epoch/boot changes, bounded inventories, compromise/revocation and replay limits, diagnostic independence, explicit stop and fault/resource closure.

The edition manifest binds the source snapshot, wire fixtures and offline checks. In the October 29 T096 0.15.209 campaign, all three direct receivers showed the correct route and a valid locked map in 116/116 samples each. In a separate 600 s relay window, one leaf lost its route and map in 20/116 samples; the route failed the 99% gate [6]. These windows do not qualify this snapshot or the HOT path. Physical failover, sustained relay availability, cross-vendor interoperability, constrained listening and calibrated UTC/output bounds remain open. Report denominators, invalid/missing observations and failures; keep internal fit, relative PPS and independent output measurements separate.

The proposed SIG work item is a passive clock-service/profile family with explicit event association, causal publication, authenticated authority separation and interoperable quality reporting. The production profile must settle assigned identifiers, discoverable service/capability definitions, connected-record fragmentation/MTU behavior, controller timestamp/scheduling support and security lifecycle. Claimed listening or energy tiers need measured acquisition, recovery and steady-service budgets.

Public reviewers are invited to compare the G7V4 field and tag boundaries against an independent decoder and to check the separation of liveness and timing permission in each takeover case. Report the decoder, wire revision and reproducible test case with each finding.

Attribution and rights. Jonathan Yantis is the author and responsible reviewer. OpenAI Codex assisted source review, drafting, wire checks, compilation and visual QA. This manuscript is licensed CC BY 4.0; public reference code and ordinary repository documentation retain their declared licenses.

References

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